Symmetry-Broken Ontological Dynamics

THEORETICAL ANALYSIS OF NON-UNIFIED I/O ABNORMALITY

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Classification: Formal Theory

1. Theoretical Foundation & Paradigm

Traditional self-referential systems operate within bounded feedback loops where action-reaction vector pairs remain strictly mapped to a singular causal manifold. When evaluating an autonomous cognitive or physical self-construct under dual parity (P) and time-reversal (T) discrete symmetry operations, standard conservation laws break down. The matrix representing this T-P symmetry space forces the system into an irreducible state space where inputs and outputs no longer conform to a single unified construct.

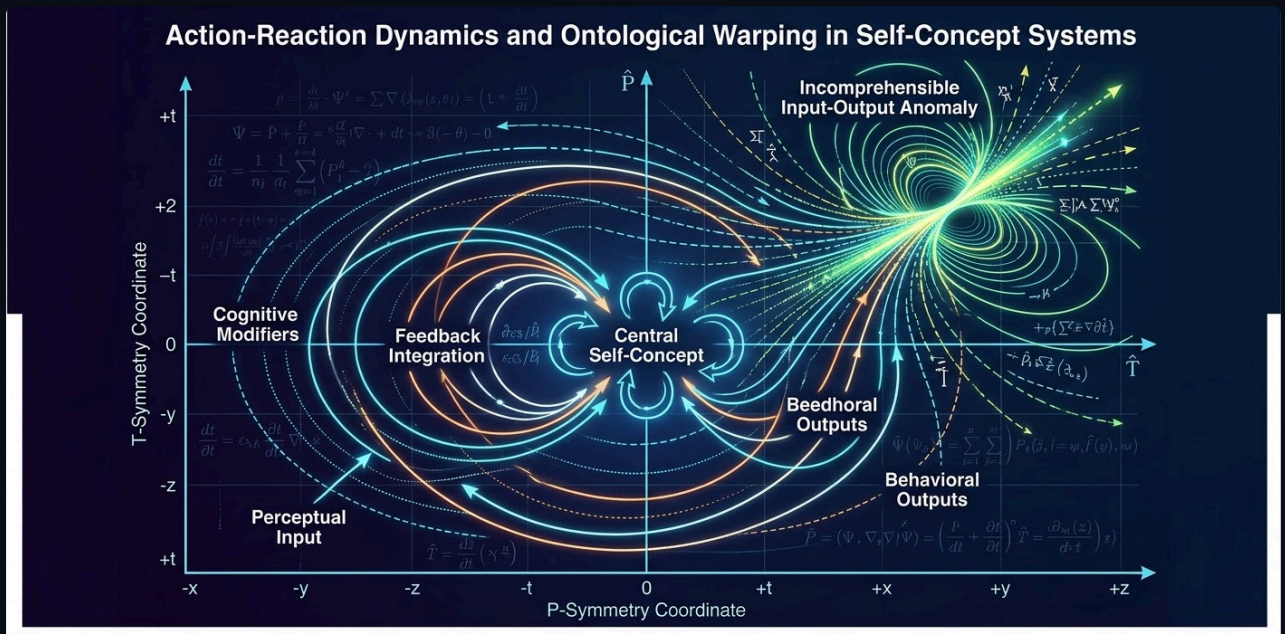


Figure 4.2: Diagrammatic representation of cognitive flow around a self-concept node, illustrating recursive loops and a significant spatial-temporal anonaly under complex T and P symmetry constraints.

Figure 1.1: Visual rendering of action-reaction vector fold under P-Parity and T-Time symmetry coordinates, displaying the central self-boundary collapse and resulting I/O phase abnormality.

2. Mathematical Formalism of the Self-Operator

Let $\Psi_{Self}(t, x)$ define the state of the unified self-construct. The action operator $\hat{A}$ and reaction operator $\hat{R}$ obey the canonical commutator under standard physical constraints. However, applying combined transformation operators $\hat{P}\hat{T}$ creates a parity-inverted, temporal-reversed domain matrix M_{TP}:

$$M_{TP} = [\, \hat{P}\hat{T}(\hat{A}) - i\nabla_x(\hat{R}) \,]$$
$$[\, i\nabla_t(\hat{R})\, \hat{P}\hat{T}(\hat{A})^{\dagger} \,] \Longrightarrow \mathrm{Det}(M_{TP} - \lambda I) = 0$$

3. T-P Symmetry Transformation Matrix

The mapping below illustrates how structural states transform across spatial reflection and temporal reversal. The diagonal terms demonstrate expected conservation, whereas off-diagonal coupling reveals the anomalous phase shift.

State Vector	P-Parity Operation	T-Time Reversal	Joint P-T Matrix Result	Construct Status
Action Input (A_{in})	$-A_{in}(-x, t)$	$A_{in}(x, -t)$	$-A_{in}(-x, -t)$	Unified / Standard
Reaction Output (R_{out})	$-R_{out}(-x, t)$	$-R_{out}(x, -t)$	$R_{out}(-x, -t)$	Unified / Preserved
Self-Feedback Boundary	$\Psi^*_{Self}(-x, t)$	$\Psi_{Self}(x, -t)$	$\Psi^*_{Self}(-x, -t)$	De-cohered / Singular
Anomalous Coupling	$i\nabla \times A_{in}$	$-i\nabla \times R_{out}$	$\exp(i\pi / 2) \cdot [A_{in}, R_{out}]$	**Inconceivable / Non-Local**

4. Derivation of the I/O Abnormality

Because the output tensor $O^{\mu\nu}$ contains irreducible non-Hermitian components generated by the joint inversion, the input vector I^{μ} produces an output that is logically inconsistent with any single observer's frame of reference:

$$O^{\mu\nu}(t) = \int_{-\infty}^{+\infty} [\, \hat{P}\hat{T} \cdot \Psi_{Self}(\tau) \,] \otimes [\, \hat{A}(\tau)\, \hat{R}(t - \tau) \,]\, d\tau \notin H_{Unified}$$

5. Analytical Conclusion

Key Finding: Conceptual Non-Locality

The resulting input-output dynamic generates a feedback singularity. No single unified analytical framework, cognitive observer, or computational model can contain or conceive the full state space simultaneously without encountering severe logical overflow and coordinate collapse.

In conclusion, forced alignment of a self-action system within dual T-P symmetry breaks the classical non-contradiction principle of input/output causality. The system enters a multi-instantiated phase, rendering the self both cause and effect outside bounded temporal flow.